A Project Report

on

Explainable Artificial Intelligence (XAI) based Intrusion Detection System

carried out as part of the ***Project Based Learning-1 (DSE2170)***

Submitted

by

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CERTIFICATE

This is to certify that the project-based learning , project titled **Explainable Artificial Intelligence (XAI) based Intrusion Detection System** is a record of the bonafide work done by **Aendrie Pawar (2430010548), Ojas Sethi (2430010550), Chhavi Gauli (2430010543)** submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Data Science) of Manipal University Jaipur, Jaipur during the academic year 2025-26.

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ABSTRACT

This study investigates Explainable Artificial Intelligence (XAI) techniques in Intrusion Detection Systems (IDS) to enhance transparency and trust in cybersecurity. It reviews and compares recent XAI-based IDS models using methods such as SHAP, LIME, and deep learning across datasets like NSL-KDD and CICIDS2017. Findings show improved interpretability and accuracy but highlight ongoing challenges in scalability and real-time deployment. This work is based on the research gaps, found through our literature review of 10 journal papers, 10 research papers, and 10 review papers.

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1. Introduction

1.1 Background

Intrusion Detection Systems (IDS) are now a part of keeping networks secure, helping detect unusual or harmful activity before it becomes a threat. With the rise in the use of machine learning (ML) and deep learning (DL), IDS have grown from simple, rule-based tools into smart systems that can detect brand-new attacks. But as these systems get smart, they also become harder to understand. That is why explainable AI (XAI) is important, as it helps make these complex models more transparent by showing why they make certain decisions, using methods like SHAP, LIME, and Grad-CAM.

After analysing 30 XAI-IDS studies, one issue keeps standing out: while XAI tools do explain model behaviour, they often don't highlight the hidden biases within the training data. These biases, whether caused by uneven class distribution, wrong labelling, or overly correlated features can push a model towards misleading predictions, even when the explanation seems to be reasonable. This mismatch between "explainable" results and actual fairness is what drives this research: to explore whether XAI can not only explain model outputs, but can also reveal the deep dataset-level biases that encourage them.

This gap between explanation and fairness motivates the present research that is to investigate whether XAI methods can not only explain model outputs but also detect their dataset-level biases that change those outputs.

1.2 Purpose of the Document

The purpose of this document is to present a improvised proposal for a research project that systematically examines the ability of XAI techniques to identify dataset-induced bias in IDS models.

It outlines:

- The rational and problem statement which was grounded in past,
- The research objectives and methodological framework,
- The experimental design that includes dataset selection, bias injection, model choice, and evaluation metrics,
- And the timeline of implementation and milestones spanning from the 2nd year to the 4th year of study.

By reading theoretical and experimental components, this proposal acts as a roadmap for a full-length empirical study that contributes to both AI transparency and cybersecurity fairness research domains.

The document acts as both:

1. A planning artifact guiding the execution of the project, and
2. A formal research proposal that demonstrates methodological rigor, literature alignment, and measurable outcomes.

1.3 Scope of the Project

The scope of this project is thought out while keeping in mind the technical feasibility, and alignment with prior XAI-IDS studies, which we reviewed.

In-Scope Components

1. **Datasets:**
    - CICIDS-2017 (primary dataset) and UNSW-NB15 (secondary dataset).
    - Controlled bias injection to our chosen datasets thus simulating real-world data imbalance.
2. **Models:**
    - Random Forest + TreeSHAP as interpretable baselines.
    - 1D-CNN + KernelSHAP/LIME as the deep-learning counterpart for black-box comparison.
3. **Evaluation Framework:**
    - Multi-level analysis based on performance, fairness metrics, and XAI quality indicators.
    - Measurement of sensitivity and bootstrap stability to evaluate robustness of XAI.
4. **Expected Contribution:**
    - Whether XAI methods can uncover biases inherited from data.

Out-of-Scope Components

- Real-time IDS deployment or online attack detection.
- Hardware-based or network-level intrusion prevention systems.
- Alternative XAI frameworks outside SHAP/LIME (e.g., Anchors, Integrated Gradients).
- Ethical or legal analysis beyond bias detection metrics.

Scope Summary

This project is for experimental bias evaluation using XAI within ML based IDS models. It does not aim to enhance model accuracy, but instead to examine the faithfulness and fairness of XAI explanations under controlled biased conditions. By this scope, our research aims to fill a critical gap found in the literature, whether explainable AI can actually pose as an auditing mechanism for data fairness in cybersecurity applications.

2. Literature Review

2.1 Overview of Existing Solutions

The integration of XAI into IDS has become a significant attention over the last decade, driven by the need for transparency and trust in black-box like IDS. Pre-existing solutions can mainly be put into two groups, one of traditional machine learning-based IDS with post- hoc explainability, and second of deep learning-based IDS enhanced with interpretability frameworks.

1. **Machine Learning-Based IDS (Tree Models and Ensembles):** Literature review work [2,4,6,9,15] primarily utilized Random Forest (RF) and Decision Trees, employing TreeSHAP or LIME to interpret feature contributions. These approaches achieved high detection accuracy while allowing analysis of important features. However, the main focus in the reviewed papers was on explanation of predictions, not on explanation validity under biased data conditions.

2. **Deep Learning-Based IDS (Neural Architectures):** Literature review work ([6,12,14,18,23]) has mentions of Convolutional Neural Networks (CNNs) and Deep Neural Networks (DNNs) for IDS, often with KernelSHAP, Grad-CAM, or Integrated Gradients. These methods showed state-of-the-art accuracy (>98%) but posed unstable or inconsistent explanations when inputs were worrying, signalling fragility in deep XAI methods.

3. **Dataset Standardization:**
   The CICIDS-2017 and UNSW-NB15 datasets are the most commonly used benchmarks in XAI-IDS literature ([2,4,6,9,10,12,15,20]) since they provide balanced traffic samples and multiple attack categories, which makes them perfect fit for reproducible experimentation.

4. **Explainability Metrics:**
   The reviewed papers mainly focus on visual or qualitative evaluation- heatmaps, feature rankings, and local explanations, without standardized metrics for fidelity, stability, or fairness. Some studies in the past ([6,14,17]) began proposing quantitative XAI measures (e.g., stability across seeds, explanation drift), but not even one of them applied them in a bias-awareness context.

Overall, existing solutions have greatly improved interpretability in IDS but have still not addressed the interplay between XAI and data bias, which remains the central research gap.

2.2 Identified Challenges

The reviewed studies provides several **serious and repetitive challenges** across XAI-based IDS research:

Challenge	**Description**	**Observed in Literature**
1. Dataset-Level Bias	Imbalanced class distributions and correlated features lead to skewed model learning.	[6,8,11,19,22]
2. Bias-Blind Explainability	XAI tools (SHAP/LIME) replicate model logic but fail to reveal biased decision boundaries.	[6,12,14,23]
3. Lack of Fairness Evaluation	Most studies emphasize accuracy without measuring false-positive/false-negative disparities across groups.	[19,22,24,25]
4. Inconsistent XAI Stability	Explanations vary across model retrains, affecting interpretability reliability.	[6,14,17]
5. Limited Metric Frameworks	Few works define standardized metrics for comparing explanation quality or faithfulness.	[4,6,15]
6. Dataset Obsolescence	Popular datasets (CICIDS-2017, UNSW-NB15) are static and may not represent evolving attack patterns.	[9,20]
7. Over-Reliance on Post-Hoc XAI	Current explainability relies on after-the-fact interpretation instead of model-intrinsic transparency.	[6,15,18]

These challenges together show that existing XAI frameworks are designed for explanation, and are not accountable; it can describe what a model does, but not whether it behaves correctly or is ethically under biased data conditions.

2.3 Opportunities for Improvement

The gaps that we have found in existing research open several opportunities for methodological and relating advancement, directly for the present study:

1. **Bias Injection and Controlled Testing:**
   By introducing artificial, measurable biases (sampling, label, covariate) in the datasets, researchers can find out if XAI techniques find or ignore these biases. This approach converts XAI from a detailed tool into a diagnostic one.
2. **Fairness-Aware Evaluation Frameworks:**
   Incorporating clarity metrics such as ΔFNR (difference in false negatives) and Equal Opportunity Gap, they allow performance to be marked not only by accuracy but also by group parity. This introduces a new layer of accountability which was not present in most studies that were done in the past.

3. **Model-Agnostic Comparison:**
   Checking both interpretable (Random Forest) and black-box (1D-CNN) models using XAI frameworks helps in identification of whether XAI blindness is model specific or universal.
4. **Quantitative XAI Quality Assessment:** Executing metrics like fidelity, stability, explanation drift, and complexity (sparsity) gives a standardized measurement system for explanation reliability and robustness.
5. **Integration of Counterfactual Testing:** Counterfactual sensitivity experiments—where a pseudo-sensitive variable is flipped (S=0 → S=1) can reveal whether explanations change meaningfully under hypothetical fairness variations.
6. **Replicability and Standardization:** Using CICIDS-2017 and UNSW-NB15 ensures reproducibility and matches with 70% of previous research, while adding bias-awareness as a novel dimension.

By addressing these opportunities, the research attempts to fill a critical empirical gap: checking whether current XAI tools can serve as effective bias-detection mechanisms in security- oriented ML systems.

2.4 Summary of Findings

The systematic literature review of 30 studies leads to the following synthesized conclusions:

Key Finding	**Implication for Proposed Research**
1. High Accuracy ≠ Fairne	IDS models can achieve excellent detection rates while remaining biased towards some traffic patterns or feature groups.
2. XAI Mirrors Model Bias	Explainability tools copies the model's reasoning without questioning biased data patterns.
3. Lack of Fairness Metrics	Existing IDS research does not integrate fairness-oriented evaluation criteria.
4. Dataset Imbalance as Hidden Driver	Most model biases come from prelying dataset structure instead of algorithmic design.
5. Unstable Deep XAI	CNN explanations (Kernel SHAP, LIME) changes across runs, undermining interpretability and reliability.
6. Methodological Gap	No study purposely injected and measured dataset bias to test XAI's bias-detection ability.

From these insights, it becomes clear that the front foot challenge is not how well XAI explains, but rather what it fails to explain—specifically, the inherited biases within the data itself.

This understanding tells the experimental design (Section 4), that proposes a bias- injection and clear and true evaluation framework to experimentally find whether XAI can identify its own blind spots in IDS applications.

This literature review provides clarity in the progress from existing XAI-IDS research to identified methodological problems and limitations and then to new research opportunities.

It strengthens the foundation for the project's core question that is:

If explainable AI techniques can detect dataset-level bias in Intrusion Detection Systems, or do they merely replicate it and position our proposal as a novel, or a literature-grounded contribution to the fields of AI clarity, transparency, and cybersecurity.

3. Problem Definition

3.1 Project Scope

The scope of this project lies at the intersection of cybersecurity, machine learning, and explainable artificial intelligence (XAI) specifically focusing on if explainability tools can detect dataset-level biases in Intrusion Detection Systems (IDS).

The project covers the conceptualization, design, implementation, and evaluation of explainable IDS models under controlled bias conditions, using benchmark datasets and standardized XAI frameworks.

In-Scope Activities

- Conducting a complte literature review of 30 peer-reviewed XAI-IDS studies to identify existing methods and gaps in the research.
- Developing an experimental design that includes:
    - Controlled bias injection (sampling, label, and covariate) to mimic realistic data skew.
    - Training and evaluation of two models:
        - Random Forest (RF) with TreeSHAP (interpretable baseline).
        - 1D-CNN with KernelSHAP/LIME (deep learning model).
- Constructing a multi-dimensional evaluation framework consisting:
    - IDS detection performance (accuracy, precision, recall, F1-score).
    - Fairness performance (ΔFNR, Equal Opportunity Gap, Group Accuracy Gap).
    - XAI quality (fidelity, stability, explanation drift, and complexity).
- Assessing whether XAI techniques remain faithful yet blind to dataset bias.
- Analysing findings and preparing a thesis and publication-ready report.

Out-of-Scope Activities

- Real-time intrusion prevention or live traffic monitoring systems.
- Use of proprietary or undisclosed network datasets.
- Implementation of non-XAI fairness algorithms or unsupervised feature debiasing methods.
- Hardware-level or cloud-based deployment of IDS tools.

This defined scope ensures the project remains academically rigorous, experimentally controlled, and methodologically grounded in the reviewed literature while focusing exclusively on bias-aware explainability in IDS.

3.2 Objectives

The project's objectives are structured to tackle the research gap discovered during the literature review, that is, XAI tools explain model outputs but do not necessarily detect dataset-level biases.

Objective Type	Description
Primary Objective	To identify whether explainable AI techniques (TreeSHAP, KernelSHAP, LIME) can detect or remain blind to dataset-induced bias in Intrusion Detection Systems.
Secondary Objectives	1. To design a controlled bias-injection framework that replicates real-world imbalances (sampling, label, covariate). 2. To evaluate how bias affects model fairness metrics and XAI explanation patterns. 3. To determine differences between interpretable (RF) and black-box (1D-CNN) models under identical bias conditions. 4. To introduce a multi-layered evaluation matrix integrating performance, fairness, and explainability. 5. To explain if explanation fidelity correlates with fairness awareness.
Exploratory Goal	To establish whether post-hoc explainability can become a data auditing mechanism for bias detection within IDS datasets.

Each objective directly aligns with gaps identified in the 30-paper reviews that have been done and contributes to advancing fairness-aware XAI research in cybersecurity contexts.

3.3 Deliverables

The following deliverables define the measurable outputs of the project over its research lifecycle (from 2nd Year, 2nd Semester to 4th Year, 2nd Semester):

Stage	Deliverable	Description	Expected Outcome
Stage 1: Literature Foundation	Comprehensive Literature Review Report	Review of 30 XAI-IDS papers, including synthesis of methods, datasets, and limitations.	Establish a theoretical base and identify research gaps.

Stage	Deliverable	Description	Expected Outcome
Stage 2: Design & Planning	Experimental Design and Methodology Chapter	Includes dataset selection, model design, bias framework, and evaluation matrix.	Approval of methodology for implementation.
Stage 3: Implementation I	RF + TreeSHAP Baseline Model	Model training and evaluation under unbiased and biased conditions.	Benchmark results for bias comparison.
Stage 4: Implementation II	1D-CNN + KernelSHAP/LIME System	Deep-learning IDS setup with XAI interpretation under bias.	Cross-model comparison for XAI blindness.
Stage 5: Analysis & Evaluation	Fairness and XAI Quality Report	Statistical analysis of fairness and explanation drift metrics.	Identification of bias-awareness patterns in XAI.
Stage 6: Documentation	Final Research Thesis & Publication Draft	Comprehensive project report with results, discussion, and recommendations.	Submission-ready thesis and research article.

Each deliverable is designed to align with semester-wise milestones (as defined in Section 5) and contribute progressively to the overall project goal.

3.4 Problem Statement

In spite of the increasing use of XAI in IDS, current research largely assumes that explainability implies fairness.
However, proof from over 30 peer-reviewed studies shows that XAI frameworks such as SHAP and LIME faithfully replicate model logic but fail to recognize underlying dataset biases for instance, when certain protocols, ports, or IP ranges are overrepresented in attack classes. As a result, models may produce technically correct but semantically biased explanations: the output appears interpretable, yet the interpretation is rooted in biased data raises a fundamental question for both AI and cybersecurity ethics: **Can XAI methods detect dataset-level bias in IDS models, or do they simply mirror the bias inherited from data?**

The absence of correct and systematic evaluation under controlled bias conditions forms the core research problem of this study.
While prior studies ([2,4,6,9,12,14,19,22]) attained high IDS accuracy and model interpretability, none explicitly tested XAI's sensitivity to dataset imbalance or feature skew. Thus, this research suggests a novel bias-injection and XAI evaluation framework that is designed to uncover whether explainability tools can serve as bias-detection mechanisms or whether they remain confined to explaining already biased reasoning.

Comparative Summary of XAI-Based Intrusion Detection JOURNAL Papers (2021–2025)

	Title	Year	Techniques Used	Dataset	Performance Metrics	Limitations Future Work	Journal Name
1	False Positive Identification in Intrusion Detection Using XAI	2023	Neural Network (NN), SHAP, Adversarial Approach	LYCOS-IDS2017	Confusion Matrix, AUC	Adds Complexity, Imperfect Filtering, Manual Step Required	IEEE Latin America Transactions, Vol. 21, No. 6, 2023
2	Robust Network Intrusion Detection Through XAI	2022	XGBoost, SHAP, Deep Autoencoder	NSL-KDD	Accuracy, Recall, Precision	Binary-only classification; limited dataset variety	IEEE Networking Letters, Vol. 4, Issue 3, 2022
3	XAI for Trust Management in IDS using Decision Tree	2021	Decision Tree (ID3)	KDD Cup 1999	Precision, Recall, F1-Score	Overfitting risk, bias in ID3	Hindawi Complexity, 2021
4	Federated XAI IDS	2025	Federated Learning, SHAP, ANN	CICIOT2023	Accuracy, Precision, Recall, F1-Score, Loss	Needs real-world testing, lower accuracy vs others	Future Internet, Vol. 17, Article 234, 2025

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5	XAI-IDS Framework for Robust Cybersecurity	2025	1D CNN, SHAP, LIME	NSL-KDD	Accuracy, Precision, Recall, F1	Limited to DDoS, static dataset	SHIFRA, Vol. 2025, pp. 69–80
6	Explainable AI for Intrusion Detection System	2022	DT, RF, SVM, Voting Classifier, LIME	CICIDS-2017	Accuracy, Precision, Recall, F1, Confusion Matrix	Extend to deep learning models, real-time analysis	Electronics, Vol. 11, Article 3079, 2022
7	Systematic Review on XAI in IDS	2025	SHAP, LIME, Rule-based, Decision Trees, Hybrid Models	CICIDS-2017, KDD Cup 99	Accuracy, FPR, Interpretability, Efficiency	Overhead from SHAP/LIME, interpretability -accuracy tradeoff	Frontiers in AI, Vol. 8, Article 1526221, 2025
8	Explainable Deep Learning IDS in IoT	2022	DNN, RuleFit, LIME, SHAP	NSL-KDD, UNSW-NB15	Accuracy, Precision, Recall, F1, TPR	Needs defense against adversarial attacks	IEEE Open Journal of Comm. Society, Vol. 3, 2022
9	Explainable Deep Learning for IoT Networks	2023	SHAP, PFI, ICE, PDP	NSL-KDD, UNSW-NB15, ToN_IoT	Accuracy, Precision, Recall, F1, Training/Detection Time	Dependent on base IDS performance, lacks vulnerability mapping	Information Sciences, Vol. 639, 2023

10	XAI-ADS for Autonomous Driving Systems	2024	DT, RF, SVM, KNN, DNN, AdaBoost, SHAP, LIME	VeReMi, Sensor Dataset	Accuracy, Precision, Recall, F1, AUC	Focus on binary classification; limited to specific datasets	IEEE Access, Vol. 12, 2024

Comparative Summary of Conference Papers on XAI-Based Intrusion Detection (2018–2025)

	Title	Year	Techniques Used	Dataset	Performance Metrics	Limitations Future Work	Conference Name
1	A framework for efficient network anomaly intrusion detection with feature selection	2018	Decision Tree, Random Forest, SVM	CICIDS2017	Accuracy: 96.25%	Limited to traditional ML models	9th International Conference on Information and Communication Systems (ICICS), 2018
2	An explainable Intelligent Intrusion Detection System	2024	Decision Tree, Random Forest, SVM	CICIDS2017	Accuracy: 96.25%	Limited generalisability / dataset scope	International Symposium on Networks, Computers and Communications (ISNCC), 2024
3	Explainable AI based Intrusion Detection in the Internet of Things	2023	XAI, ML and Deep Learning Models	CIC IoT Dataset 2022, IEC 60870-5-104 Intrusion Detection Dataset	Accuracy (DT): 99.85%	Real-world heterogeneity, evolving attack patterns, adversarial behavior	18th International Conference on Availability, Reliability and Security (ARES 2023)
4	X-CBA: Explainability Aided CatBoosted	2024	XAI, Taxonomy building,	N/A	N/A	Lack of consensus/standardization, evaluation gap, adaptation issues	IEEE International Conference on

	Anomal-E for Intrusion Detection System		Literature collection				Communications (ICC), 2024
5	Evaluating Explainable AI for Deep Learning-based Network Intrusion Detection System Alert Classification	2025	XAI, Deep Learning Models	CSCAS	N/A	Does not deeply evaluate resource usage	11th International Conference on Information Systems Security and Privacy (ICISSP 2025)
6	Explainable Machine Learning for Computer Security Applications	2023	XAI	N/A	N/A	Lack of stakeholder focus, weak integration into workflow	IEEE 8th European Symposium on Security and Privacy (EuroS&P), 2023
7	An Explainable Intelligent Intrusion Detection System on SPBO with Explainable AI	2023	SPBO, Random Forest, ML, XAI	NSL-KDD	Accuracy: 99.81%	Single dataset, generalization, explainability overhead	IEEE Asia Pacific Conference on Computer Science and Data Engineering (CSDE), 2023
8	An Explainable Intelligent Intrusion Detection System	2024	LinearSVC, SGD Classifier, Decision Tree, Random Forest	CICIDS2017	Accuracy: 99%	Limited scope, practical live systems, potential overfitting	International Symposium on Networks, Computers and Communications (ISNCC), 2024

Comparative Summary of REVIEW/RESEARCH Papers

	Title	Year	Techniques / Approach Used	Dataset	Performance Metrics	Limitations / Future Work	Source
1	A systematic review on the integration of explainable artificial intelligence in intrusion detection systems to enhancing transparency and interpretability in cybersecurity.	2025	SHAP, LIME, rule-based and hybrid XAI–IDS models using PRISMA-based review.	NSL-KDD, CICIDS2017, UNSW-NB15	Detection Accuracy, Precision, Recall, F1-Score, FPR, TPR, AUC, computational efficiency	Trade-off between accuracy & interpretability, outdated datasets, lack of standard metrics, high computational cost. Future: lightweight real-time XAI, standard metrics, adaptive IDS.	A Systematic Review on the Integration of XAI in IDS, 2025
2	Evaluating machine learning-based intrusion detection systems with explainable AI: enhancing transparency and interpretability	2025	ML models (Decision Tree), XAI techniques: SHAP, preprocessing via feature selection and normalization.	UNSW-NB15	Accuracy, Precision, Recall, F1, ROC-AUC, FPR=0.07, FNR=0.12	Typical ML-XAI trade-offs. Future: apply to complex datasets, advance XAI for broader cybersecurity.	Frontiers in Computer Science, 2025

3	A Novel Ensemble of Deep Learning Approach for Cybersecurity Intrusion Detection with Explainable AI	2025	Hybrid ensemble: ANN + SVM (base) with Random Forest meta-classifier.	NSL-KDD (22,544 records, 41 features)	Accuracy, Precision, Recall, F1, ROC-AUC	Binary classification only. Future: extend to multiclass attacks.	Journal Report, 2025
4	Use of Explainable Artificial Intelligence for Analyzing and Investigating Cybercrime in Digital Forensics	2025	XGBoost, TabNet (deep model) for forensic/cybercrime data.	UNSW-NB15	Feature-importance evaluation, local explanation stability.	Tree vs DL models show varied consistency. Future: improve explanation stability and forensic transparency.	Computers Journal, Vol. 14, Issue 5, Article 160, 2025
5	A Literature Review on Applications of Explainable Artificial Intelligence (XAI)	2025	Model-agnostic explainers (LIME, SHAP), model-specific (Integrated Gradients, Grad-CAM, LRP).	Multiple domain-specific datasets.	Accuracy, AUC, fidelity, stability, explanation usability, computational cost.	Lack of standard benchmarks; trade-offs between interpretability & performance. Future: standardization.	IEEE Access Journal, 2025
6	Intrusion detection in the Internet of Things using convolutional neural networks: an explainable AI approach	2025	Lightweight 1D-CNN model.	TON-IoT	Accuracy 0.995, Precision 0.9949, Recall 0.9947, F1 0.9947	Deployment challenges on IoT devices. Future: optimize for real-time & privacy.	Cybersecurity (SpringerOpen), 2025

7	Enhancing Threat Detection and Response using Explainable AI (XAI): A Literature Review	2025	Survey of XAI techniques for cybersecurity.	Various prior studies.	Summarized metrics from prior works (accuracy, FPR, interpretability).	Limited scalability; lack of standardized metrics. Future: modular XAI frameworks.	Journal Article, 2025
8	Intrusion Detection Using Hybrid Random Forest and Attention Models and Explainable AI Visualization	2025	Hybrid Random Forest + Attention-based model.	Not clearly specified (public preview)	No numeric metrics provided.	Future: explore XAI opportunities, challenges, and research gaps.	Journal of Internet Services and Information Security (JISIS), 2025
9	Explainable artificial intelligence for botnet detection in Internet of Things	2025	Rule extraction & distillation, LIME, SHAP.	N-BaIoT (IoT dataset)	Qualitative effectiveness summary.	Challenges: black-box complexity, IoT variability. Future: real-time deployment.	Scientific Reports (Nature), 2025
10	Evaluating explainable AI for deep learning-based network intrusion detection system alert classification	2025	LSTM model; XAI methods: LIME, SHAP, Integrated Gradients, DeepLIFT.	SOC dataset from TalTech University (real-world)	Faithfulness, complexity, robustness, reliability.	Transparency vs trust challenge. Future: generalization & analyst feedback integration.	ICISSP 2025 Proceedings

4. Proposal Development

4.1 Research Findings

Our research is built upon the findings derived from 30 reviewed studies on XAI based IDS, including 10 journal papers, 10 research papers, and 10 conference papers.

Across these works, a consistent pattern arises: while post-hoc explainability tools such as TreeSHAP, KernelSHAP, and LIME accurately mimicked the reasoning of IDS, they still failed to expose the underlying bias present in the datasets.

Studies including [2,4,6,9,12,14] showed that Random Forest and CNN-based IDS had greater accuracy, yet their XAI outputs remained "technically correct but semantically blind" meaning that they reflected learned model behaviour without detecting the dataset-level skew. This recurring limitation in the literature shaped the foundation of our proposal: to systematically test whether XAI tools can detect bias inherited from data, rather than simply rationalizing the model's predictions.

The review, furthermore, revealed that over 70% of XAI-IDS studies depend on benchmark datasets such as CICIDS-2017 and UNSW-NB15, given their balanced feature sets, public availability, and alignment with existing SHAP/LIME baselines. However, none of the reviewed papers explored controlled bias conditions, a gap our study aims to address.

4.2 Project Objectives

Based on our literature review, the core and subsidiary objectives are formulated as such:

Objective Type	**Description**
Primary Objective	To evaluate if the XAI techniques (TreeSHAP, KernelSHAP, LIME) can detect dataset-induced bias in IDS.
Secondary Objectives	1. To inject controlled bias (sampling, label, and covariate) into IDS datasets and measure its impact on fairness and XAI outputs. 2. To record explainability quality (fidelity, stability, explanation drift) across traditional (RF) and deep (1D-CNN) IDS models. 3. To propose a multi-dimensional evaluation framework including IDS accuracy, fairness. 4. To check if "bias blindness" is model-agnostic across different architectures.

Each objective of our traces back to the identified literature gaps from our literature review.

4.3 Methodology

Our study hopes to follow an experimental design including dataset bias manipulation, model training, and XAI-based evaluation. The overall rationale, dataset choice, and technique selection are directly grounded in literature to make sure methodological alignment and comparability.

4.3.1 Data Collection Methods

Two benchmark datasets are selected for the study:

Dataset	**Reason for Selection**	**Key Features**	**Advantages**	**Disadvantages**
CICIDS-2017	Most widely cited dataset in IDS + XAI literature ([2,4,6,9,10,12]).	80+ features, multiple attack families.	Realistic, feature-rich, and standardized.	Slight class imbalance; older dataset.
UNSW-NB15	Used in explainability-focused IDS studies ([4,6,15,20]).	49 mixed-type features across 9 attack types.	Modern design, reproducibility.	Synthetic traffic; lacks real-time variance.

Bias Injection Framework :

To replicate realistic bias while maintaining control, a three-step framework is employed:

Step	**Technique**	**Description**	**Rationale from Literature**
1. Pseudo-sensitive Variable (S)	Derived from network attributes (e.g., port range, subnet, protocol).	Introduces controllable but plausible grouping.	Inspired by correlations noted in [6,13,19,25].
2. Bias Injection Methods	Sampling bias, label bias, covariate shift.	Creates controlled imbalance scenarios.	Motivated by imbalance findings in [8,11,22].
3. Unbiased Test Set	Independent clean set for evaluation.	Isolates performance vs bias effects.	Follows data-isolation setups in [4,6,9].

By introducing synthetic yet measurable bias, we can explore whether XAI can uncover or overlook such distortions.

4.3.2 Analytical Techniques

Model Architectures

Model	**Description**	**Justification**	**Supporting Studies**
Random Forest (RF)	Ensemble model interpreted via TreeSHAP.	Stable, interpretable, standard XAI benchmark.	[2,4,6,9,15]
1D-CNN (Lightweight)	Sequential convolution over features.	Tests XAI performance on deep architectures.	[6,12,14,18,23]

1D-CNN Layers (Adapted from [12,14])

- Conv1D(64, k=3) → Local feature extraction
- BatchNorm + ReLU → Stabilization
- MaxPooling(2) → Dimensionality reduction
- Conv1D(128, k=3) → Deep synthesis
- GlobalAvgPooling → Prevents overfitting
- Dense(128, ReLU) + Dropout(0.3) → Regularization
- Output → Softmax/Sigmoid classification

This lightweight CNN mirrors prior IDS structures but improves interpretability integration.

Explainability Methods

XAI Tool	**Model**	**Reason for Selection**	**Cited Literature**
TreeSHAP	Random Forest	High fidelity and reproducibility.	[2,4,6,9,15]
KernelSHAP	CNN	Model-agnostic SHAP variant.	[6,12,14]
LIME (Optional)	CNN	Provides linear surrogate validation.	[4,6,14]

These choices align with 73% of prior XAI-IDS research, ensuring comparability and methodological rigor.

Evaluation Framework:

Four performance categories are used:

Category	Metrics	Purpose / Literature Basis
IDS Detection	Accuracy, Precision, Recall, F1, ROC-AUC	Baseline performance ([2,4,6,10,14]).
Fairness	ΔFNR, ΔFPR, Equal Opportunity Gap, Group Accuracy Gap	Measures bias impact ([19,22,24,25]).
XAI Quality	Fidelity, Stability, Explanation Drift, Complexity, Cost	Evaluates explainability robustness ([6,12,14,17]).
Other Metrics	Counterfactual Sensitivity, Bootstrap Stability, Resource Usage	Causal consistency and efficiency ([6,14,20]).

This multi-layered matrix makes sure balanced evaluation of accuracy, fairness, and interpretability addressing an evident gap in prior studies.

4.4 Expected Outcomes

Aspect	Expected Observation	Supporting Literature	Interpretation
Baseline (Unbiased Data)	High accuracy, stable SHAP explanations.	[2,4,6]	Confirms XAI fidelity under clean conditions.
Post-Bias Injection	Accuracy remains high, but fairness gaps widen.	[19,22]	Indicates hidden model bias.
XAI Outputs	Explanations remain “technically correct” yet ignore bias.	[6,14]	Demonstrates XAI’s inability to detect inherited bias.

Hence, XAI tools are expected to remain truthful but bias-blind, supporting the hypothesis that the present explainability frameworks justify decisions without auditing data-level bias. The proposed study explicitly extends present literature by treating bias as an experimental variable testing if XAI systems can recognize, rather than reproduce, the biases they inherit from data.

5. Timeline and Milestones

Overview

The project is designed over a three-year research trajectory beginning in the 2nd Year, 2nd Semester and concluding in the 4th Year, 2nd Semester.
Each semester contributes a distinct and progressive research milestone starting from conceptual foundation and literature review to experimentation, analysis, and final thesis submission.

All milestones match with the objectives and methodology that is established in Section 4, ensuring growth sequentially, academic grounding, and traced progress toward our study's ultimate goal:

To evaluate whether XAI techniques can find out or remain blind to dataset-level bias in Intrusion Detection Systems (IDS).

Detailed Semester-Wise Plan

Academic Year & Semester	**Duration**	**Major Activities & Deliverables**	**Milestones / Expected Outcomes**
2nd Year – 2nd Semester	Months 1–6	Foundational Stage • Conduct detailed literature review of 30 XAI-IDS papers, focusing on methodologies, datasets, and explainability limitations .• Identify consistent trends in IDS transparency and bias. • Draft research gap analysis. • Finalize problem statement, objectives, and rationale.• Develop a preliminary proposal framework (as seen in "Proposal Development").	Literature review completed. 30-paper synthesis mapped. Research gap clearly defined. Proposal document drafted.
3rd Year – 1st Semester	Months 7–12	Design & Planning Stage • Formulate experimental design based on reviewed studies. • Select datasets,bias injection framework, models and XAI tools and justify why the chosen are chosen. • Write a methodology section with reasoning.	Experimental Design finalized. Dataset pipeline defined. Bias injection methods designed. Methodology chapter approved.

Academic Year & Semester	Duration	Major Activities & Deliverables	Milestones / Expected Outcomes
3rd Year,2 Semester	Months 13–18	Implementation Stage (Phase I) • Start dataset preprocessing and feature analysis .• Implement Random Forest + TreeSHAP stack (baseline setup). • Inject controlled biases into dataset (sampling, label, covariate) .• Conduct initial fairness metric evaluation (ΔFNR, ΔFPR, group accuracy gap). • Record baseline vs biased model differences.	Biased datasets constructed. Random Forest baseline tested. Preliminary fairness gap observed. Midterm review submission.
4th Year – 1st Semester	Months 19–24	Implementation Stage (Phase II) • Develop 1D-CNN (lightweight) model and integrate KernelSHAP/LIME explainability. • Perform training and testing on unbiased and biased datasets. • Compute XAI Quality Metrics (Fidelity, Stability, Explanation Drift, Complexity). • Compare model-XAI behavior across both architectures.	CNN implementation complete. All explainability metrics computed. XAI drift under bias documented. Cross-model comparison achieved.
4th Year – 2nd Semester	Months 25–30	Analysis, Documentation & Publication Stage • Analyze performance, fairness, and XAI interpretability trends. • Synthesize expected outcomes vs. observed results. • Draft thesis chapters • Conduct final presentation and defense.	Final results validated. Thesis chapters finalized. Research paper submitted. Project completed and defended.

Milestone Summary

Stage	Semester	Key Deliverable	Indicator of Completion
Stage 1: Foundation	2nd Year – 2nd Sem	30-paper Literature Review + Proposal Draft	Submission of finalized research proposal.
Stage 2: Design Setup	3rd Year – 1st Sem	Experimental Design & Methodology Finalization	Approved "Experimental Design & Methodology" chapter.
Stage 3: Implementation I	3rd Year – 2nd Sem	Bias Framework + Random Forest Evaluation	Working biased datasets + baseline metrics.
Stage 4: Implementation II	4th Year – 1st Sem	CNN + KernelSHAP Integration	XAI quality evaluation completed.
Stage 5: Analysis & Reporting	4th Year – 2nd Sem	Final Thesis & Publication	Completed thesis and defense presentation.

5.1 Summary

Our research proposal presents a plan to look into the domain of Explainable Artificial Intelligence (XAI) and cybersecurity, while questioning if the XAI techniques, so depended upon, detect dataset-level bias in Intrusion Detection Systems (IDS), or do they merely justify it.
Building upon insights from 30 reviewed studies, our literature review established that majority of the XAI-based IDS models achieve high interpretability and accuracy but lack fairness-awareness**.** While tools such as TreeSHAP**,** KernelSHAP**,** and LIME can effectively visualize model reasoning, prior works indicate they often fail to recognize biases emerging from imbalanced, skewed, or correlated features within IDS datasets.
To address this gap, our proposal defines a structured experimental design featuring:

- Controlled bias injection (sampling, label, and covariate) in widely used datasets **(**CICIDS-2017**,** UNSW-NB15);
- Comparative evaluation between Random Forest (interpretable baseline) and 1D-CNN (black-box) models;
- Integration of XAI tools (TreeSHAP**,** KernelSHAP**,** and LIME) to measure fidelity, stability, explanation drift, and fairness metrics (ΔFNR, Equal Opportunity Gap).

The timeline and milestones then outline a 3-year progression from literature review and methodology design in the 2nd year to implementation, evaluation, and thesis submission by the 4th year.
By combining explainability, fairness, and data-bias evaluation, this project aims to contribute a novel fairness-aware perspective to the growing field of XAI-driven IDS research.
Overall, the study enhances the understanding that explainability does not automatically ensure fairness and aims to experimentally determine whether XAI can evolve from an interpretive tool into a diagnostic mechanism for bias detection in cybersecurity systems.

5.2 Next Steps

Moving ahead, the following steps will be undertaken to transform this proposal into an executable research project:

Phase	**Planned Activities**	**Expected Outcome**
1. Preliminary Setup (Current Stage)	Finalize research proposal, secure departmental approval, and identify required computational resources (datasets, libraries, and GPU access).	Approved project proposal and resource allocation.
2. Implementation Preparation	Begin dataset acquisition and preprocessing (CICIDS-2017, UNSW-NB15). Develop initial scripts for bias injection and model setup**.**	Working data pipeline and baseline model framework.
3. Experimental Execution	Train and evaluate Random Forest and 1D-CNN under biased and unbiased scenarios using SHAP and LIME. Collect performance and fairness metrics.	Empirical results quantifying XAI's bias awareness.
4. Analysis and Validation	Conduct cross-model comparison and fairness–explainability correlation analysis**.** Validate results with multiple seeds and explainability consistency checks.	Verified findings and refined hypothesis.
5. Documentation and Dissemination	Compile results, write thesis chapters, and prepare conference/journal publication based on findings.	Final research thesis and scholarly dissemination.

The project will thus proceed systematically from conceptual groundwork to empirical validation**,** ensuring traceability across all phases. The next immediate step involves finalizing the methodology for departmental review**,** followed by dataset preparation and baseline implementation in the upcoming semester.

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